

Lec. 05

Vector Properties

RANK: $\rightarrow$ matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

$\underbrace{\quad \quad \quad}_{\bar{a}_1} \quad \underbrace{\quad \quad \quad}_{\bar{a}_2} \quad \underbrace{\quad \quad \quad}_{\bar{a}_n}$

n columns of size m .

$\rightarrow m \times 1$.

COLUMN SPACE.

$$\begin{aligned} \bar{y} &= A \bar{x} \\ &= [\bar{a}_1 \ \bar{a}_2 \ \dots \ \bar{a}_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \end{aligned}$$

$$= \bar{a}_1 x_1 + \bar{a}_2 x_2 + \dots + \bar{a}_n x_n$$

Linear combination of columns
of A .

$$y \in C(A)$$

↓
column space
of A .

$C(A)$ = subspace spanned
by columns of A .

i.e. it contains all possible
linear combinations of columns
of A .

LINEAR INDEPENDENCE :

$\bar{a}_1, \bar{a}_2, \dots, \bar{a}_n$
linearly independent

if $x_1 \bar{a}_1 + x_2 \bar{a}_2 + \dots + x_n \bar{a}_n = 0$

$\Leftrightarrow x_1 = x_2 = \dots = x_n = 0$.

linearly
independent

There do NOT exist scalars

$x_1, x_2, \dots, x_n$, NOT all zero

such that $x_1 \bar{a}_1 + \dots + x_n \bar{a}_n = 0$.

$$\text{Ex: } \bar{a}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \quad \bar{a}_2 = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$$

$$\bar{a}_3 = \begin{bmatrix} 7 \\ -1 \\ -3 \end{bmatrix}$$

Are $\bar{a}_1, \bar{a}_2, \bar{a}_3$ linearly independent?

$$\begin{aligned} 3\bar{a}_1 + 2\bar{a}_2 &= 3 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} \\ &= \begin{bmatrix} 7 \\ -1 \\ -3 \end{bmatrix} = \bar{a}_3 \end{aligned}$$

$$3\bar{a}_1 + 2\bar{a}_2 = \bar{a}_3$$

$$\Rightarrow 3\bar{a}_1 + 2\bar{a}_2 - \bar{a}_3 = 0$$

$\uparrow \quad \quad \uparrow \quad \quad \uparrow$
 $x_1=3 \quad x_2=2 \quad x_3=-1$

$\therefore \bar{a}_1, \bar{a}_2, \bar{a}_3$: NOT linearly independent.

linearly dependent.

$\bar{a}_1, \bar{a}_2$: These are linearly independent

i.e. there are NO scalars

x_1, x_2 NOT both zero,
Such that

$$\underline{x_1 \overline{a_1} + x_2 \overline{a_2} = 0.}$$

RANK OF MATRIX

Rank of matrix (A)

column rank $\left\{ \begin{array}{l} = \text{maximum number of linearly} \\ \text{independent columns.} \end{array} \right.$

row rank $\left\{ \begin{array}{l} = \text{maximum number of} \\ \text{linearly independent rows.} \end{array} \right.$

For any matrix A,
column rank = row rank = rank (A)
 $\leq \min(m, n)$

How to determine rank?

Row operations
Gaussian elimination.

$$A = [\bar{a}_1 \quad \bar{a}_2 \quad \bar{a}_3]$$

$$= \begin{bmatrix} 1 & 2 & 7 \\ -1 & 1 & -1 \\ 1 & -3 & -3 \end{bmatrix} \quad \text{3x3}$$

$$\text{rank} \leq \min(3, 3)$$

$$\therefore \text{rank} \leq 3.$$

$$A = \begin{bmatrix} 1 & 2 & 7 \\ -1 & 1 & -1 \\ 1 & -3 & -3 \end{bmatrix} \begin{matrix} \rightarrow R_1 \\ \rightarrow R_2 \\ \rightarrow R_3 \end{matrix}$$

$$R_2 \leftarrow R_2 + R_1 \quad \begin{bmatrix} 1 & 2 & 7 \\ 0 & 3 & 6 \\ 1 & -3 & -3 \end{bmatrix}$$

$$R_2 \leftarrow \frac{R_2}{3} \quad \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 1 & -3 & -3 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - R_1 \quad \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 0 & -5 & -10 \end{bmatrix}$$

Row Echelon
form

pivots
↑↑

$R_3 \leftarrow R_3 + 5R_2$

$$\begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

2 non-zero rows
 $\therefore \text{rank} = 2$

3rd row completely zero!

$\Rightarrow$ maximum # of
linearly independent columns = 2.
= max. no. of linearly indep. rows.